Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18CE31]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

FLUID MECHANICS

Time: 3 Hours **Maximum Marks: 70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Explain different types of fluids and draw the relation between shear stress and velocity gradient for these fluids. **07**

b. Determine the intensity of shear of an oil having viscosity of 1 poise. The oil is used for lubricating the clearance between a shaft of diameter 10 cm and its journal bearing. The clearance is 1.5 mm and the shaft rotates at 150 r.p.m **07**

2. a. What are the gauge pressure and absolute pressure at a point 3 m below the free surface of a liquid having a density of 1.53×10^3 kg/m^3 if the atmospheric pressure is equivalent to 750 mm of mercury, the specific gravity of mercury is 13.6 and density of water is 1000 kg/m^3. **06**

b. A differential manometer is connected at the two points A and B of two pipes as shown in Fig 1. The pipe A contains a liquid of specific gravity of 1.5 while pipe B contains a liquid of specific gravity of 0.9. The pressures at A and B are 1 kgf/cm^2 and 1.8 kgf/cm^2 respectively. Find the difference in mercury level in the differential manometer. **08**

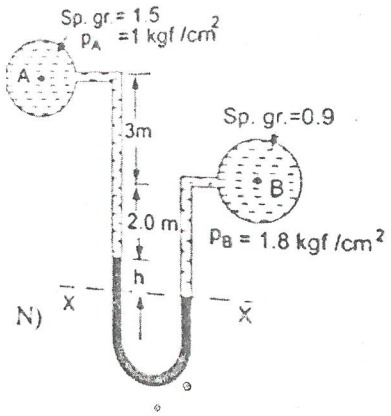

Fig 1

3. a. A 30 cm diameter pipe, conveying water, branches into two pipes of diameters 20 cm and 15 cm respectively as shown in Fig.2. If the average velocity in the 30cm diameter pipe is 2.5 m/s, find the discharge in this pipe. Also determine the velocity in the 15 cm pipe if average velocity in the 20 cm diameter pipe is **06**

2 m/s.

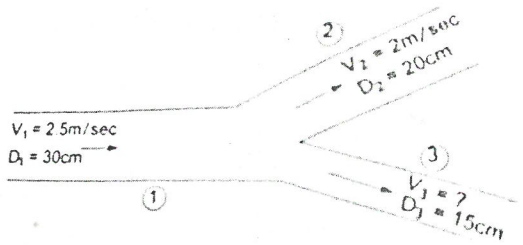

Fig.2

b. Differentiate between

- **i.** Steady and unsteady flow
- **ii.** Uniform and non-uniform flow **08**
- **iii.** Rotational and irrotational flow
- **iv.** Compressible and incompressible flow

OR

4. a. Derive the continuity equation in rectangular coordinates for three- dimensions fluid element. **06**

b. The velocity potential function is given by an expression

$\emptyset = \frac{-xy^3}{3} - x^2 + \frac{x^3 y}{3} + y^2$ **08**

i. find the velocity components in x and y directions

ii. show that ø represents a possible case of flow

5. a. Derive Euler's equation of motion. Obtain Bernoulli's equation from Euler's equation. **08**

b. A pipe, through which water is flowing, is having diameters, 20 cm and 10 cm at the cross sections 1 and 2 respectively. The velocity of water at section 1 is given 4m/s. find the velocity head at sections 1 and 2 and also rate of discharge. **06**

OR

6. a. The water is flowing through a pipe having diameters 20 cm and 10 cm at sections 1 and 2 respectively. The rate of flow through the pipe is 35 lit/sec. the section 1 is 6mabove datum and section 2 is 4m above datum. If the pressure at section 1 is 39.24 N/cm^2 find the intensity of pressure at section 2. **07**

b. The water is flowing through a taper pipe of length 100 m having diameters 600mm at the upper end and 300mm at the lower end, at the rate of 50 liters/sec. the pipe has a slope of 1 in 30. Find the pressure at the lower end, if the pressure at the higher level is 19.62 N/cm^2 **07**

7. a. The head of water over an orifice of diameter 100 mm is 10 m. The water coming out from orifice is collected in a circular tank of diameter 1.5 m. the rise of water in this tank 1.0 m in 25 sec. also the coordinates of a point on the jet, measured from vena-contracta are 4.3 m horizontal and 0.5 m vertical. Find the coefficients C_d, C_v, C_c **10**

b. Find an expression for discharge through large rectangular orifice. **04**

OR

8. a. Calculate an expression for discharge over a rectangular notch. **06**

b. Find the discharge over a triangular notch of angle 60^0 when the head over the V notch is 0.3m. assume C_d =0.6 **08**